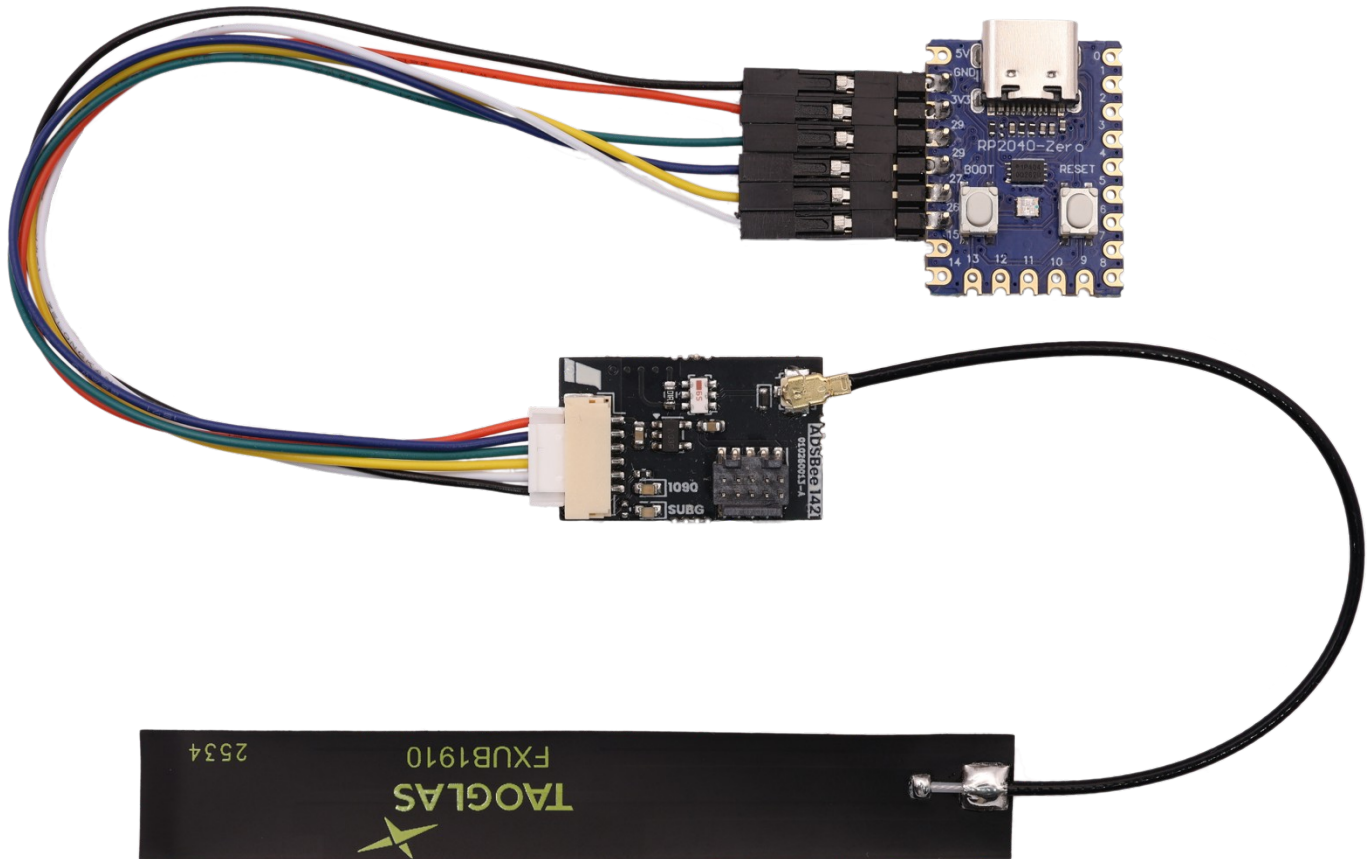




ADSBee 1421 Devkit

Ultra Low-Power Dual Band ADS-B Receiver Module
Programming + Evaluation Kit





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Revision History

20260806

- Initial release.



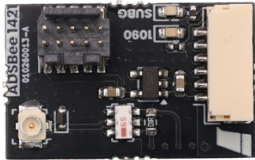
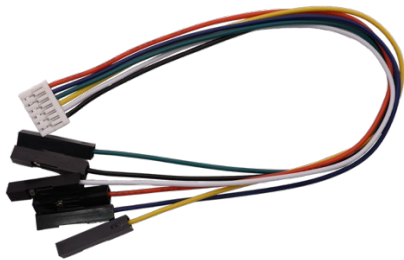
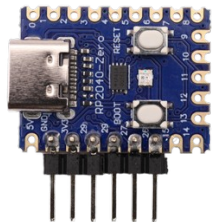
1 Overview

This kit contains everything you need to get started developing with ADSBee m1421! The included PCBA is a variant of the ADSBee 1421 with an included 10-pin ARM SWD programming header, allowing breakpoint debugging on the ADSBee m1421 module's CC1314 RF microcontroller using a compatible debug probe (Segger J-Link or equivalent probe that supports cJTAG is recommended).

The kit also includes a wire harness that breaks out the ADSBee 1421's 6-pin JST-GH header to 0.1" DuPont pin headers, allowing it to be connected to arbitrary interface devices, or the included ADSBee 1421 Programmer board.



2 Kit Contents

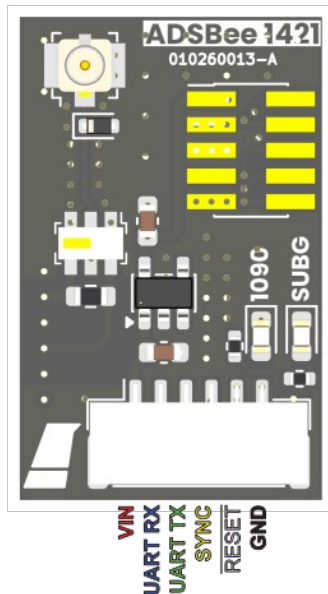
Part Image	PN	Qty	Description
ADSBee 1421 Devkit PCBA 	010260017	1	Dual-band ADS-B receiver devkit. Same as the ADSBee 1421, but includes a 10-pin ARM SWD header to allow breakpoint debugging via cJTAG.
Taoglas FXUB1910 1090/978MHz Dual-band Flex Antenna 	070260005	1	Dual-band ADS-B flex antenna.
ADSBee 1421 JST-GH to DuPont Pin Socket Breakout 	050260006	1	Cable for connecting ADSBee 1421 to programmer.
ADSBee 1421 Programmer PCBA 	010260018	1	ADSBee 1421 programmer and serial bridge.



3 Device Pinouts

3.1 ADSBee 1421 Devkit PCBA

ADSBee 1421 Connector Pinouts



	1	2	
VCC	•	•	SWDIO / TMS
GND	•	•	SWDCLK / TCK
GND	•	•	SWO / TDO
KEY	•	•	NC / TDI
GNDDetect	•	•	nRESET
	9	10	

10-PIN ARM SWD CONNECTOR

All lines are TTL logic levels (0-VCC).

Connects to CC1314R10 via cJTAG or 4-wire JTAG.

Includes VCC sense and GND.

COMMS CONNECTOR

All lines are TTL logic levels (0-VCC).

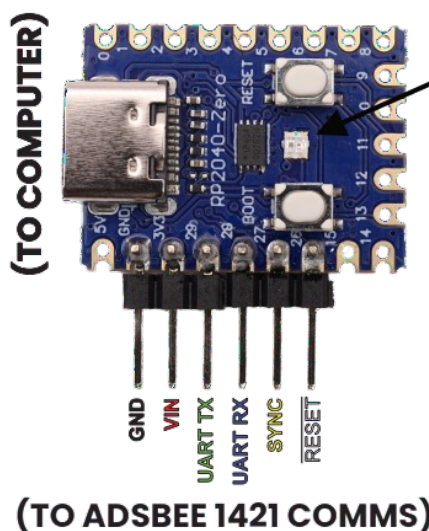
VIN range is 1.8-5.5VDC.

Hold SYNC high on RESET rising edge to enter UART bootloader.

UART interface is "CONSOLE" and can be used for control commands, reporting protocols, and logging.

3.2 ADSBee Programmer

ADSBee 1421 Programmer



LED Color

- yellow blink: waiting for device
- orange: forced reflash armed
- cyan: CRC check before flash
- magenta blink: flashing
- blue blink: post-flash CRC verify
- blue: console baud negotiation
- green: console pass-through
- red: error

press BOOT to start flashing a 1421

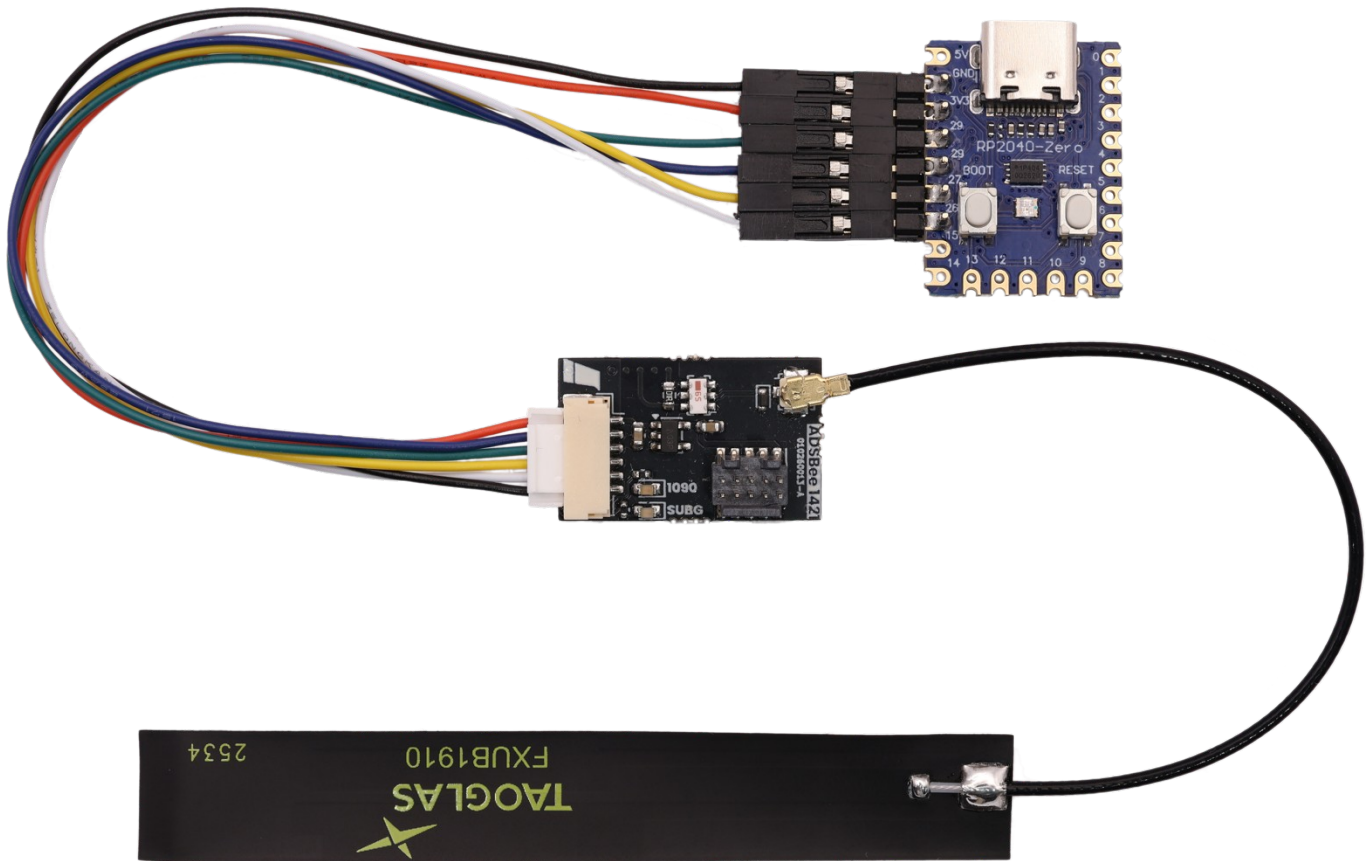
hold BOOT and press RESET to load new firmware image from PC



4 Evaluation

4.1 Connection Diagram

For programming and evaluation, connect ADSBee 1421 to the ADSBee 1421 programmer as shown below.



The ADSBee 1421 Programmer will enumerate as a USB CDC device called “ADSBee”. Once the light on the ADSBee 1421 Programmer turns green, it functions as a serial bridge and connects the ADSBee 1421 to the computer. A standard UART to USB converter can be substituted if desired.

Note that the default baud rate on the ADSBee 1421 module is 1000000 baud. Changing the baud rate from this value could result in the module being unable to communicate via the ADSBee 1421 Programmer.



4.2 Aircraft Map

A live aircraft map is available from adsbee.aero/1421-console. Use the yellow “Connect” button on the top right corner of the webpage to connect to your ADSBee 1421 via web serial, then click on the “Map” tab to see aircraft you are receiving. The HTML page may also be downloaded for local use (it’s just a single HTML file)!

ADSBee 1421

Console

Map

Disconnect

Interpolate positions

ICAO

CALLSIGN

TYPE

SQWK

ALT (FT) ↓

SPD (KT)

HDG

LAT

LON

RSSI

aaed0c

ASA278

adsb_icao

37,000

444

161°

36.9008

-120.4929

-32768.0

ad8ed4

ASA1397

adsb_icao

35,350

490

333°

35.9044

-121.1138

-32768.0

ac57af

adsb_icao

35,050

460

174°

-32768.0

ab349c

ASA1417

adsb_icao

35,000

489

62°

37.4920

-122.4767

-32768.0

780f77

CSN327

adsb_icao

33,000

480

118°

36.6202

-122.2838

-32768.0

ac1789

SWA4302

adsb_icao

29,825

457

140°

36.5059

-122.0612

-32768.0

a174f7

SKW3086

adsb_icao

28,500

443

300°

37.1142

-121.0544

-32768.0

ac5a94

SWA4532

adsb_icao

21,550

376

146°

36.9756

-122.5398

-32768.0

abc14d

adsb_icao

18,300

379

318°

-32768.0

a6757a

N515RA

adsb_icao

17,300

361

136°

37.3594

-122.3174

-32768.0

abb5e7

SKW5986

adsb_icao

17,125

379

48°

-32768.0

a611db

UAL2478

adsb_icao

17,000

403

77°

37.9356

-121.4130

-32768.0

a4dfd2

FFT2638

adsb_icao

15,775

409

129°

37.2816

-122.1520

-32768.0

a37765

SKW3937

adsb_icao

13,550

372

356°

37.4693

-121.9525

-32768.0

aaaddc

SWA178

adsb_icao

13,275

356

317°

37.1940

-121.3288

-32768.0

acb1e5

adsb_icao

12,550

332

262°

37.5696

-121.4691

-32768.0

a8fc73

UAL1702

adsb_icao

12,250

368

76°

37.8381

-121.8578

-32768.0

a6ce36

ASA634

adsb_icao

9,550

255

224°

37.5354

-121.6721

-32768.0

a254d7

UAL927

adsb_icao

8,925

250

172°

37.4405

-122.2686

-32768.0

a5a2d9

UPS943

adsb_icao

8,550

296

143°

37.2915

-121.7452

-32768.0

a9ff49

adsb_icao

8,150

292

180°

37.5624

-122.5766

-32768.0

a1f5ff

UAL810

adsb_icao

7,875

240

353°

37.2241

-122.3939

-32768.0

a44212

UAL2841

adsb_icao

7,400

245

271°

37.4194

-121.8155

-32768.0

ac390f

AAL3138

adsb_icao

6,550

234

42°

37.3355

-122.2362

-32768.0

a06a09

AAL2333

adsb_icao

5,600

227

298°

37.4732

-122.0245

-32768.0

ab5dd4

UAL436

adsb_icao

4,375

201

298°

37.5183

-122.1307

-32768.0

SWA3585

Status

Airborne

ICAO

ac0bb9

Callsign

SWA3585

Type

adsb_icao

Category

A3

Alt baro

3,350 ft

Alt geom

3,375 ft

Speed

245 kt

Track

43.0°

Vert rate

1,280 fpm

Latitude

37.44740

Longitude

-121.96356

RSSI

-32768.0 dBm

Messages

1

NIC

8

NIC baro

1

NAC p

10

NAC v

0

SIL

3

GVA

2

SDA

2

ADS-B version

2

Last seen

1 s ago



5 Updating Firmware

The included ADSBee 1421 Programmer board began its life as a humble RP2040 Zero module, but with the addition of some pin headers and firmware, has been transformed into a serial bridge and firmware updater compatible with any ADSBee 1421 / m1421 device.

5.1 Updating an ADSBee 1421 / m1421

Each ADSBee 1421 Programmer includes a ADSBee 1421 firmware built into its flash, and when connected to an ADSBee 1421 / m1421 device over UART (with SYNC and RESET also connected), it will automatically connect to the attached ADSBee and bring its firmware up to date. Once the firmware update is complete, the ADSBee 1421 Programmer will function as a USB to serial converter, allowing the user to interact with the ADSBee 1421 / m1421 via USB.

5.2 Updating an ADSBee Programmer

To update the firmware image baked into an ADSBee 1421 Programmer, follow the steps below.

- 1) Download the latest firmware release from the releases page in the ADSBee github repository (<https://github.com/CoolNamesAllTaken/adsbee/releases>). The firmware file will be labelled "adsbee_1421_programmer-X.X.X.uf2", where X.X.X is the firmware version of the baked-in firmware image (e.g. 0.3.6), and uf2 is the filetype.
- 2) Connect the ADSBee 1421 Programmer to your computer via USB. Hold the BOOT button and click RESET; the ADSBee 1421 Programmer will show up as a flash drive labelled RPI-RP2. Let go of both buttons.
- 3) Drag and drop the adsbee_1421_programmer-X.X.X.uf2 firmware file onto the RPI-RP2 drive and allow the transfer to complete. The ADSBee 1421 Programmer now has a new firmware and is ready to update ADSBee 1421 / m1421 devices via UART.